

ASSIGNMENT #03

Course: CHE 433: PROCESS SIMULATION WITH ASPEN PLUS
Term: Fall 2025
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Due Date: Friday November 14th, my office (6:00PM)

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name(s) in the report.
- Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.
- Upload your Aspen Plus Simulation file (*.apw)
- Bring a hard copy of your report as a PDF file by the Due Date and Time in class

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

Problem 1: Biomass Pyrolysis Simulation

Biomass conversion processes that are based on pyrolysis offer high yields of liquid transportation fuels such as gasoline and diesel blend stocks, or refinery intermediates. Fast pyrolysis proceeds by heating the biomass to an intermediate temperature where volatilization begins but the gasification of vapor-phase products does not occur. Fast pyrolysis generates a liquid bio-oil that can be transported efficiently and upgraded to fuels.

In a conceptual process design (see Process Flowsheet Diagram Figure 1. below), the fast pyrolysis reactor system is usually represented as a dual circulating fluidized bed. Biomass and hot sand are introduced into the bottom of a vertical riser reactor. As the solids are carried up the riser by a fluidizing gas, the rapid heating of biomass particles to around 450-500 degrees Celsius, produces non-condensable gases and vapors of oxygenated hydrocarbons. Unreacted biomass flows out as solid char. Solid and vapor phases exit the top of the riser and are separated in a cyclone. Sand and char drop into a char combustor, which is basically a second fluidized bed where the unreacted char is combusted in air to reheat the sand. The energy to drive the endothermic pyrolysis reactions is thus provided via the hot sand, by partial combustion of the biomass itself. The vapor phase is condensed downstream to form the bio-oil. In some cases, the bio-oil separates into organic and aqueous phases. Fast pyrolysis bio-oils can be as high as 50% oxygen by mass and are generally assumed to be hydro-treated downstream, to reduce oxygen.

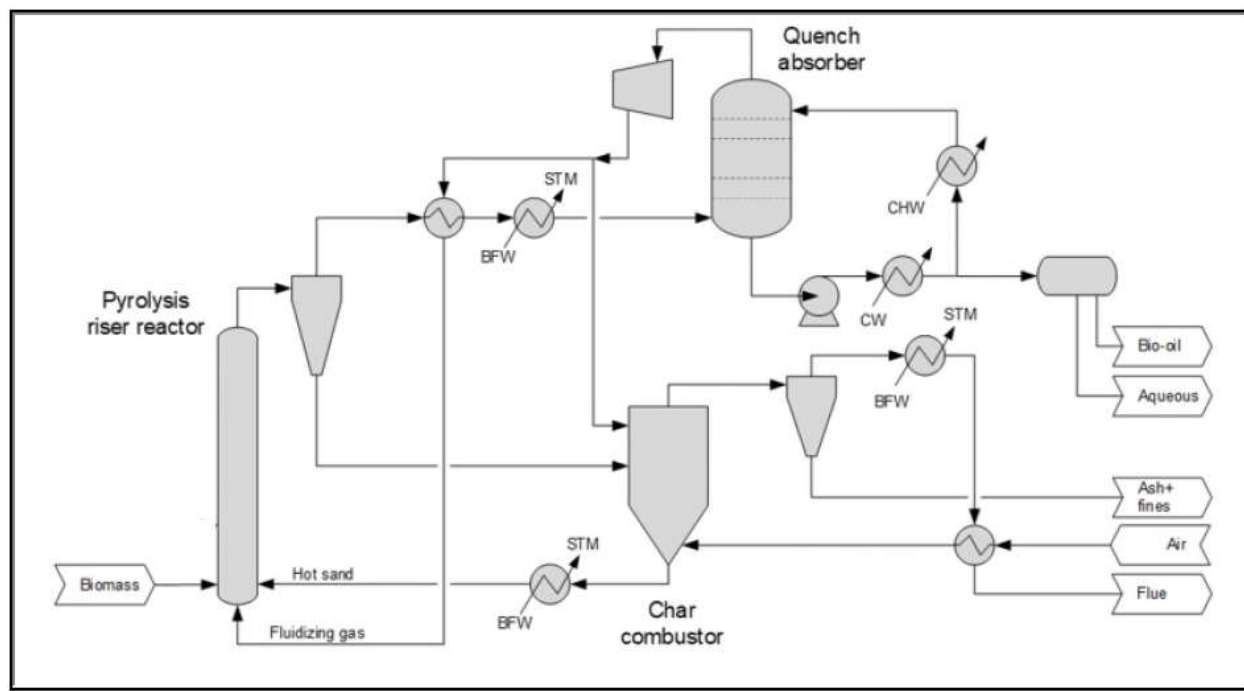


Figure 1: Process Flow Diagram(PFD) of Biomass Fast Pyrolysis

The objective of the assignment is to prepare an ASPEN Plus simulation and calculate:

1. Total Power(KW) consumed
2. Total High(250 °C) and Low(125 °C) pressure steam generated (MMSCFD)
3. Mass yield (%) of incoming carbon to **Bio-oil**
4. Mass yield (%) of incoming carbon to **Flue Gas**
5. Mass yield (%) of incoming carbon to **Waste Water**

Your final report should also include the following

1. Material and Energy Balance with the UIC template
2. A brief outline of your methodology for modeling the pyrolysis reactor
3. A brief outline of your methodology for modeling the char combustor

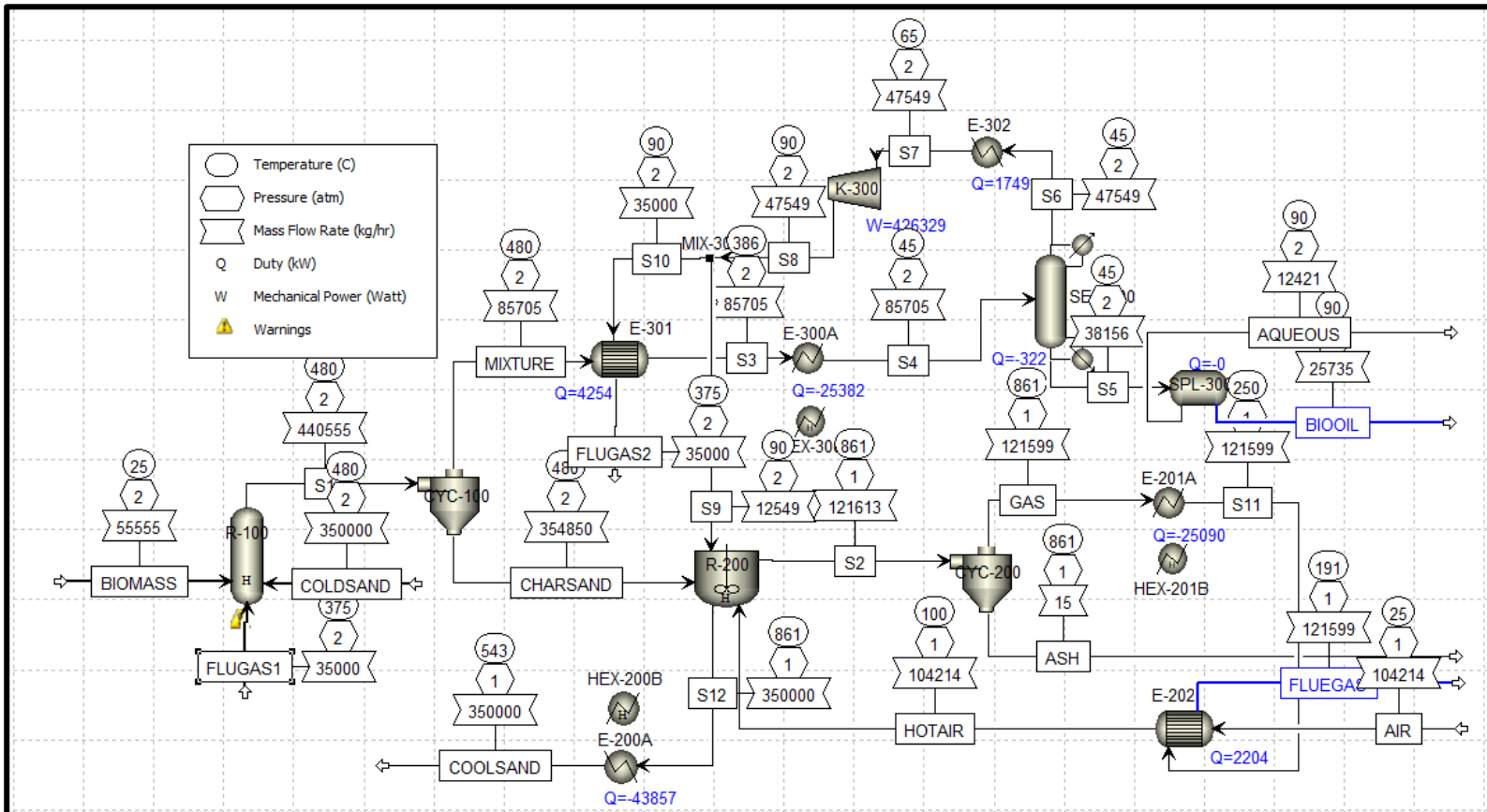
The Quench Absorber can be simulated using a SEP2 Aspen Plus Unit Operation block

Data for the simulation

1. Bone dry biomass flow rate: **50,000 kg/hr**
2. Biomass feed temperature: **25 deg C**
3. Biomass feed pressure: **2.3 atm**
4. Moisture weight fraction: **0.1**
5. Fluidization gas to Sand mass ratio: **0.1**
6. Bleed Gas to Combustor: **13,000 kg/hr**
7. Bleed Gas temperature: **80 deg C**
8. Bleed Gas pressure: **2.31 atm**
9. Sand to dry biomass mass ratio: **7.0**
10. Biomass **PROXANAL**
 - a. Moisture:
 - b. Fixed Carbon (FC): $0.85*(100-ASH)$
 - c. Volatile Matter (VM): $0.15*(100-ASH)$
 - d. Ash (ASH): 0.3000
11. Biomass **ULTANAL**
 - e. ASH: 0.30
 - f. CARBON: 49.66
 - g. HYDROGEN: 6.31
 - h. NITROGEN: 0.10
 - i. CHLORINE: 0.00
 - j. SULFUR: 0.08
 - k. OXYGEN: 43.55
12. Biomass **SULFANAL**
 - l. PYRITIC: 0.00
 - m. SULFATE: 0.04
 - n. ORGANIC: 0.04

13. Pyrolysis Reactor Pressure Drop: 0.3 atm
14. Pyrolysis Reactor Temperature: 480 deg C
15. Pyrolysis Reactor Yield (from Experiments)
- o. Gas: **12.9%**
 - i. H₂: 0.10%
 - ii. CO: 5.82%
 - iii. CO₂: 6.03%
 - iv. CH₄: 0.53%
 - v. C₂H₄: 0.42%
 - p. Water: **14.8%**
 - q. Char: **9.7%**
 - r. Organics: **62.6%**
 - i. Acids[Acetic Acid]: 11.1%
 - ii. Carbonyls[Acetone]: 12.7%
 - iii. Phenols[m-Cresol]: 4.0%
 - iv. Methoxyphenols[Guaiacol]: 27.5%
 - v. Levoglucosan: 4.4%
 - vi. Furfural: 2.9%
16. Combustor Pressure: **1.12 atm**
17. Combustion products: **NO₂, SO₂, CO₂, H₂O**
18. Oxygen in the combustor effluent: **2% mol**
19. Fluidization Gas temperature: **375 deg C**
20. Fluidization Gas pressure: **2.31 atm**
21. Fluidization Gas and Bleed Gas composition, **mol%**
- s. CO: 31.19
 - t. CO₂: 20.11
 - u. Water: 15.09
 - v. NO₂: 0.30
 - w. SO₂: 0.91
 - x. Methane: 2.62
 - y. Ethylene: 1.51
 - z. Acetone: 28.27
22. Sand component: SiO₂
23. Temperature of the Quench Absorber (SEP2): 45 deg C
24. Pressure of the Quench Absorber: 1.7 atm

Section 1: Process Flowsheet Diagram (PFD) in Landscape mode displaying Temperature, Pressure, Heat/Work and Mass Flow.



Process Flow Sheet

Stream Name	Units	S5	S6	S7	S8	S9	S10	S11	S12
Description									
From		SEP-300	SEP-300	E-302	K-300	MIX-300	MIX-300	E-201A	R-200
To		SPL-300	E-302	K-300	MIX-300	R-200	E-301	E-202	E-200A
Stream Class		MIXCINC	MIXCINC	MIXCINC	MIXCINC	MIXCINC	MIXCINC	MIXCINC	MIXCINC
Enthalpy Flow	kW	-81106.1557	-74510.18517	-72761.44972	-72335.1206	-19090.13625	-53244.98378	-99944.68023	-1378204.984
Mass Flows	kg/hr	38156.31052	47548.68948	47548.68948	47548.68948	12548.68948	35000	121598.6344	350000
O2	kg/hr	0	0	0	0	0	0	2640.747274	0
N2	kg/hr	0	0	0	0	0	0	79946.01729	0
H2	kg/hr	0	50	50	50	13.19562076	36.80437924	0	0
CO	kg/hr	0	10895.88761	10895.88761	10895.88761	2875.560014	8020.327592	0	0
CO2	kg/hr	0	11105.02705	11105.02705	11105.02705	2930.75451	8174.272539	29539.6607	0
H2O	kg/hr	12351.96508	3087.99127	3087.99127	3087.99127	814.9592343	2273.032036	9290.522122	0
NO2	kg/hr	0	126.1594815	126.1594815	126.1594815	33.29505347	92.86442805	33.29505347	0
SO2	kg/hr	0	532.9044847	532.9044847	532.9044847	140.6401096	392.2643751	148.3919784	0
CH4	kg/hr	0	649.2101961	649.2101961	649.2101961	171.3346308	477.8755653	0	0
C2H4	kg/hr	0	597.2188873	597.2188873	597.2188873	157.613479	439.6054083	0	0
TOLUENE	kg/hr	0	0	0	0	0	0	0	0
PXYLENE	kg/hr	0	0	0	0	0	0	0	0
BENZENE	kg/hr	0	0	0	0	0	0	0	0
MNAPHTHEN	kg/hr	0	0	0	0	0	0	0	0
INDENE	kg/hr	0	0	0	0	0	0	0	0
INDANE	kg/hr	0	0	0	0	0	0	0	0
ACETONE	kg/hr	854.3454379	20504.29051	20504.29051	20504.29051	5411.336831	15092.95368	0	0
CONALDE	kg/hr	0	0	0	0	0	0	0	0
CYPENTON	kg/hr	0	0	0	0	0	0	0	0
2MEFURAN	kg/hr	0	0	0	0	0	0	0	0
BZFUAN	kg/hr	0	0	0	0	0	0	0	0
FURFURAL	kg/hr	1450	0	0	0	0	0	0	0
5MEFURAN	kg/hr	0	0	0	0	0	0	0	0
MCRESOL	kg/hr	2000	0	0	0	0	0	0	0
AACID	kg/hr	5550	0	0	0	0	0	0	0
ETMOXPHE	kg/hr	0	0	0	0	0	0	0	0
GUAIACOL	kg/hr	13750	0	0	0	0	0	0	0
NAPHTHOL	kg/hr	0	0	0	0	0	0	0	0
LVG	kg/hr	2200	0	0	0	0	0	0	0
SAND	kg/hr	0	0	0	0	0	0	0	350000
ASH	kg/hr	0	0	0	0	0	0	0	0
WOOD	kg/hr	0	0	0	0	0	0	0	0
CHAR	kg/hr	0	0	0	0	0	0	0	0
C	kg/hr	0	0	0	0	0	0	0	0
SULFUR	kg/hr	0	0	0	0	0	0	0	0

Section 3: Solution Logic and Description

The objective of this problem is to extract bio-oil from wood through fast pyrolysis of biomass. After the bio-oil has been extracted, it could be converted to fuel through hydro-treatment to reduce the oxygen content. Below, the solution process is shown to undergo this process which mainly involves biomass pyrolysis, char combustion, and bio-oil separation.

Methodology of Modeling Pyrolysis Reactor:

In order to model the pyrolysis reaction of this process, moisture must be removed from the biomass to produce the yields given in the problem. First the feed stream of biomass is inputted with the temperature being 25 C, pressure is 2.3 atm, and the following Proxanal, Ultanal, and Sulfanal are inputted all in the Nonconventional tab of the stream:

Biomass PROXANAL	
a. Moisture:	
b. Fixed Carbon (FC):	$0.85 * (100 - \text{ASH})$
c. Volatile Matter (VM):	$0.15 * (100 - \text{ASH})$
d. Ash (ASH):	0.3000
Biomass ULTANAL	
e. ASH:	0.30
f. CARBON:	49.66
g. HYDROGEN:	6.31
h. NITROGEN:	0.10
i. CHLORINE:	0.00
j. SULFUR:	0.08
k. OXYGEN:	43.55
Biomass SULFANAL	
l. PYRITIC:	0.00
m. SULFATE:	0.04
n. ORGANIC:	0.04

Proxanal, Ultanal, and Sulfanal of Biomass

Also 50,000 kg/hr was added as a flow rate for wood, additionally knowing that there is 10% moisture in the biomass, through percent addition and using excel the amount of moisture in the stream was found and inputted in the conventional components tab to be 5555.55 kg/hr

The moisture in the biomass can be removed through passing it first in a dryer where air will be supplied to heat up the biomass from 25 C to 125 C, with this all the moisture is removed. Next, a Splitter is added to separate the moisture from the biomass. The dried biomass is then cooled back to 25 C using a cooler to allow for optimal reaction conditions. Now the biomass is ready to be fed into the pyrolysis reactor where the following yields are input along with passing hot sand as an inert:

o. Gas:	12.9%
i. H ₂ :	0.10%
ii. CO:	5.82%
iii. CO ₂ :	6.03%
iv. CH ₄ :	0.53%
v. C ₂ H ₄ :	0.42%
p. Water:	14.8%
q. Char:	9.7%
r. Organics:	62.6%
i. Acids[Acetic Acid]:	11.1%
ii. Carbonyls[Acetone]:	12.7%
iii. Phenols[m-Cresol]:	4.0%
iv. Methoxyphenols[Guaiacol]:	27.5%
v. Levoglucosan:	4.4%
vi. Furfural:	2.9%

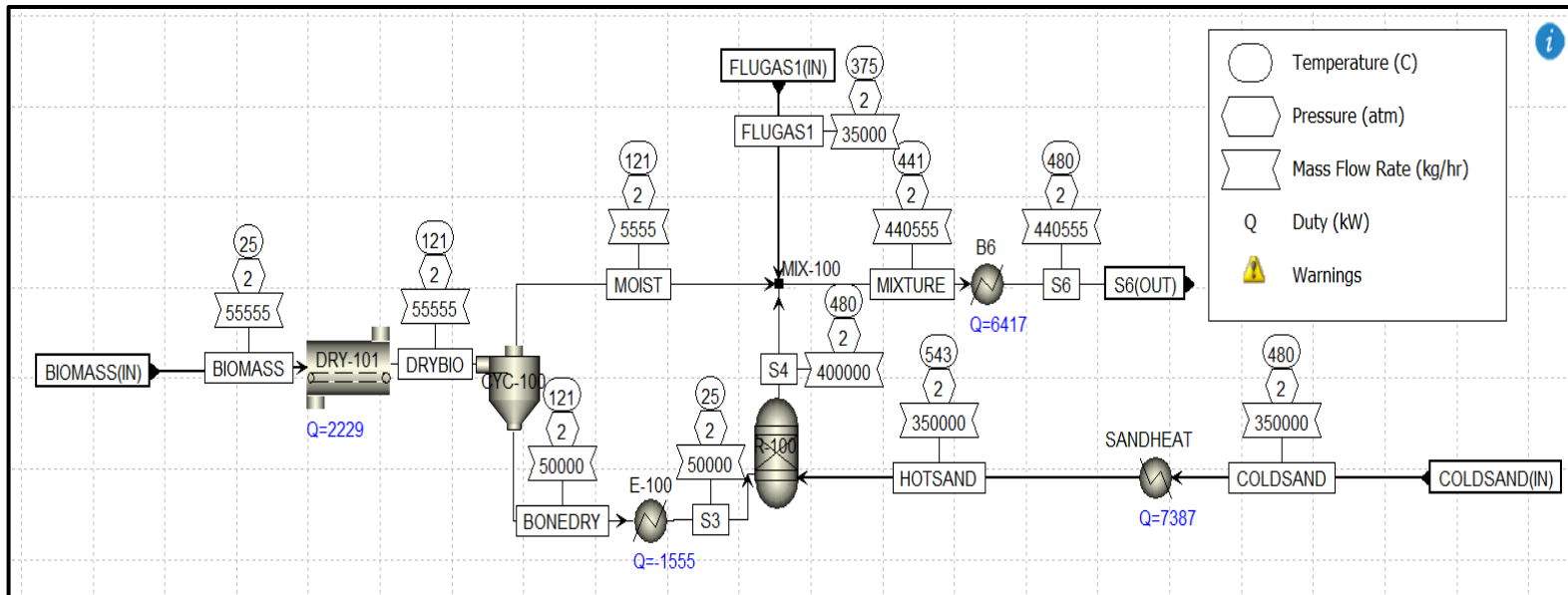
Yields from Experiments

Another thing that must be inputted is the reaction temperature which is 480 C along with inputting component attributes for the non-conventional components that will be produced from the reactor. For Ash: the only input in the Ultanal would be 100% Ash, for the Proxanal 100% Ash, and no input for the Sulfanal. For Char the Ultanal, Proxanal, and Sulfanal are the same as the wood which can also be passed as component attribute in the reactor.

The Sand is fed with an unknown temperature, since the pyrolysis reactor is modeled as an adiabatic process, the temperature of the sand can be found through creating a design specification in the simulation. The specification is to set the duty (QCalc) of the reactor to equal zero by varying the temperature of the cooler that the sand is leaving from within a range from 450-550 C. The temperature of the sand leaving the cooler was found to be 543 C.

The reactor effluent, flue gas stream, and the splitted moisture stream are then mixed and put into a heater to raise the temperature to 480 C.

The following was all done in a hierarchy and considered fake operations and don't resemble reality but was performed to perform the simulation of this process, the hierarchy is attached below:



Pyrolysis Reactor Hierarchy

Although modeling this reaction was successful, the simulation did run with warnings due to the atom balance not being satisfied. An attempt was made to satisfy the atom balance using a solver to adjust the reaction yield, which was unsuccessful. With that being said, the reaction ran and gave results that are realistic and match with past results.

Methodology of Modeling Bio-oil Separation:

After the pyrolysis reaction occurred, the effluent is a mixture of solids and vapors. The solids, which include sand and char are separated from the vapor mixture using a solid splitter and then transferred to a char combustor to reheat and recycle the sand, generate flue gas, and dispense ash and solid fines. The vapors are sent to a heat exchanger which is cooled strictly from the cold upstream recycled flue gas specified so the flue gas exits at 375 C. The vapor mixture is then sent to a condenser to cool the mixture to 45 C to push the stream in a saturated state in the quench absorber.

The quench absorber which is modeled as a SEP2 distillation column, is input with the following split fractions for the downstream products:

O2	Split fraction	1
N2	Split fraction	1
H2	Split fraction	1
CO	Split fraction	1
CO2	Split fraction	1
H2O	Split fraction	0.2
NO2	Split fraction	1
SO2	Split fraction	1
CH4	Split fraction	1
C2H4	Split fraction	1
TOLUENE	Split fraction	0
PXYLENE	Split fraction	0
BENZENE	Split fraction	0
MNAPTHEN	Split fraction	0
INDENE	Split fraction	0
INDANE	Split fraction	0
ACETONE	Split fraction	0.96
CONALDE	Split fraction	0
CYPENTON	Split fraction	0
2MEFURAN	Split fraction	0
BZFURAN	Split fraction	0
FURFURAL	Split fraction	0
5MEFURAN	Split fraction	0
MCRESOL	Split fraction	0
AACID	Split fraction	0
ETMOXPHE	Split fraction	0
GUAICOL	Split fraction	0
NAPHTHOL	Split fraction	0
LVG	Split fraction	0
SAND	Split fraction	0
C	Split fraction	0
SULFUR	Split fraction	0
AIR	Split fraction	0

Split Fractions for Quench Absorber

The upstream products which include flue gas, and small amounts of unseparated bio-oils is sent to a compressor. Before sending the upstream products to the compressor, the stream was sent to a heater to heat the products to 65 C which was a temperature found via trial and error and was set in place to remove the error in the simulation regarding flashpoint and liquid being in the compressor. The heater was able to eliminate this error through the vaporization of the remaining liquid.

The downstream products which include bio-oils and water are sent to a decanter to split the organic phase of the bio-oils from the aqueous phase of water. The decanter was specified to separate the water with 100% efficiency, allowing for a near complete separation of the bio-oil and water. The figure below illustrates the steps taken to process the downstream and upstream products:

Methodology for Modeling Char Combustor

Modeling the Char Combustor requires the utilization of two R-Stoic reactors in series and a pseudo-reaction to break down the char into its components. The incoming stream to the first R-stoic is the outlet stream of the first cyclone, CYC-100, containing only char and sand. A combustion reaction takes organic materials and burns them into CO_2 , H_2O , and other components based on the reactant stream makeup. Because char is a nonconventional component, there is no reaction stoichiometry to model its breakdown, which includes the CO_2 , H_2O , and another nonconventional component, ash. To bypass this, a pseudo-reaction within the R-Stoic reactor is utilized. Because char is burnt biomass, its makeup can be described the same way – using the ultimate analysis (ULTANAL) of the biomass. Thus, within the R-Stoic, we can select to have all the char be converted into its components:

Edit Stoichiometry

Reaction No. 1

Component	Coefficient
CHAR	-1

Component	Coefficient
ASH	1
C (CSOLID)	1
H2	1
N2	1
SULFUR	1
O2	1

Products generation

☐ Molar extent kmol/hr

☒ Fractional conversion of component CHAR

Close

Here the components are taken to be ash, solid carbon (graphite), hydrogen gas, nitrogen gas, solid sulfur and oxygen gas. The coefficients of the reaction are initially all taken to be 1, and the fractional conversion of char is taken to be 1 (100%). To find what the true stoichiometric coefficients are, a calculation is done by defining the ULTANAL and taking the weight fraction of each component on a Wet Basis and dividing that by the molecular weight of each component:

$$\text{Stoichiometry} = \frac{\text{weight fraction on wet basis}}{\text{molecular weight of comp. } i} \rightarrow \gamma = \frac{w_i}{MW_i}$$

A calculator block is created to calculate and export these values into the reactor:

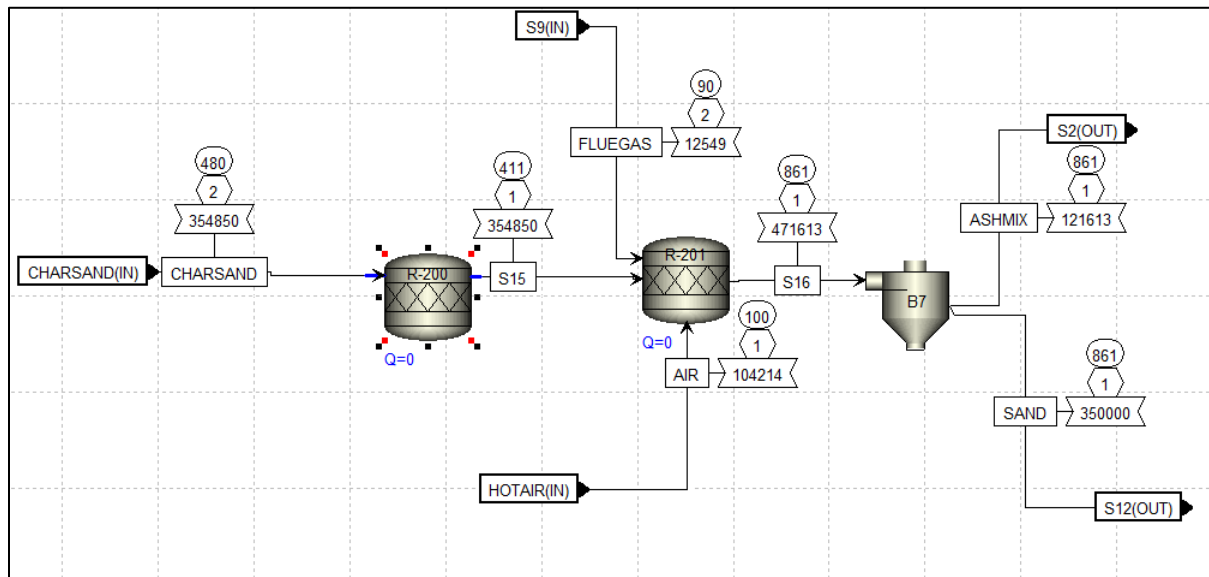
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W0= MOIST/100
DRY=1-W0
NO2=DRY*ULTIMATE (7) /MO2/100.0
NN2=DRY*ULTIMATE (4) /MN2/100.0
NASH=DRY*ULTIMATE (1) /100.0
NC=DRY*ULTIMATE (2) /MC/100.0
NH2=DRY*ULTIMATE (3) /MH2/100.0
NS=DRY*ULTIMATE (6) /MS/100.0

```

This completes the first R-Stoic reactor.

Now that the components of char are created, the actual combustion of these components must now be modeled. Very simply, a second R-Stoic can be set to “Combustion” and an inflow of air to supply oxygen is added to complete this reaction. Additionally, recycled gas from the Quench Absorber is inserted into the second R-Stoic to increase yields. The flowsheet looks as follows:



Here, the flowrate of air, HOTAIR(IN) is determined by setting a design spec on the mole fraction of oxygen in the outlet of the reactor to equal 0.02. Additionally, both reactors are ran at 1.2 atm, as specified in the problem.

Section 4: Miscellaneous Results

Calculations			
	Name	Units	Value
▶	Total Power	KW	426.329
▶	Bio-Oil Yield	UNITLESS	62.2801
▶	Flue Gas Yield	UNITLESS	32.4683
▶	Waste Water Yield	UNITLESS	0.151448
▶	High Pressure Ste...	MMSCFD	33.908
▶	Low Pressure Stea...	MMSCFD	37.3419

Final Results

Total Power Calculation

The Total Power was taken as the brake power required to set the effluent stream of the Quench Absorber, coming out at a pressure of 1.7atm, to 2.31 atm in the compressor. This value came out to be 426.329kW.

Mass Yield Calculations

To calculate the Bio-Oil yield, the amount of carbon inside the Biomass stream and the Bio-Oil stream needed to be found. The carbon inside the biomass stream was found by multiplying the Bone-Dry basis of 50,000kg/hr by the ULTANAL of carbon (49.66%). This results in the total flowrate of carbon inside the biomass. To find the carbon inside the bio-oil product stream, the weight fraction of carbon was found inside of each component and then multiplied by the mass flow rate of that component. The sum of these flowrates resulted in the total flowrate of carbon inside the bio-oil product stream. Lastly, the flowrate of carbon inside the bio-oil stream is divided by the flowrate of carbon inside the biomass and multiplied by 100 to get the yield %. The same analysis was taken for the yield of Flue Gas and Waste Water. The calculations can be seen below:

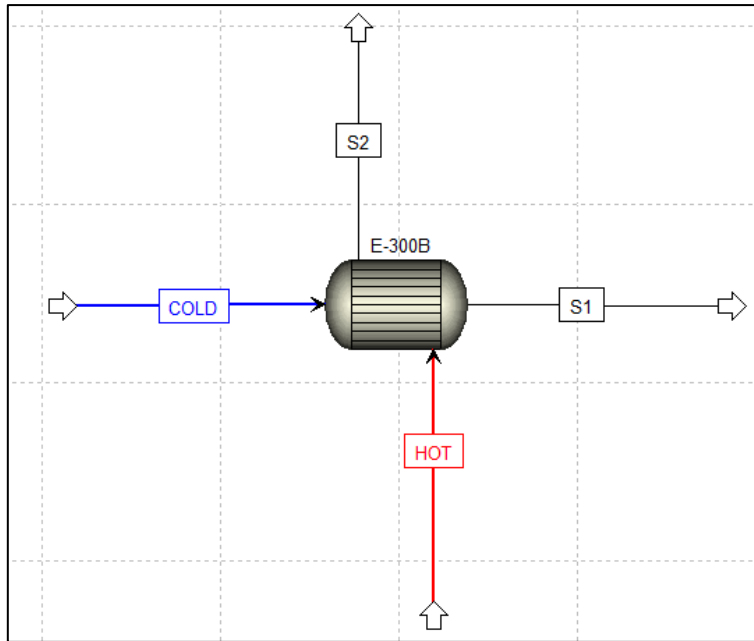
Compound	Compound Flow (kg/hr)	Carbon Frac.	C flow (kg/hr)
Acetone	853.95	0.62	529.79
AA	5503.82	0.40	2201.62
Furfural	1443.90	0.63	902.46
m-cresol	1999.62	0.78	1554.67
Guaiacol	13739.69	0.68	9305.61
Levoglucosan	2182.45	0.44	970.01
Total C flow (kg/hr)	Biomass C Flow (kg/hr)	Mass Yield (%)	
15464.158	24830.000	62.280	Bio-Oil

Compound	compound flow (kg/hr)	carbon fraction	C flow (kg/hr)
CO2	29539.762	0.273	8061.888
Total C flow (kg/hr)	Biomass C Flow (kg/hr)	Mass Yield (%)	
8061.888	24830.000	32.468	Flue Gas

Compound	compound flow (kg/hr)	carbon fraction	C flow (kg/hr)
Acetone	0.394	0.620	0.244
AA	46.175	0.400	18.471
Furfural	6.102	0.625	3.814
m-cresol	0.383	0.777	0.298
Guaiacol	10.303	0.677	6.978
Levoglucosan	17.549	0.444	7.800
Total C flow (kg/hr)	Biomass C Flow (kg/hr)	Mass Yield (%)	
37.604	24830.000	0.151	Wastewater

Steam Generation Calculations

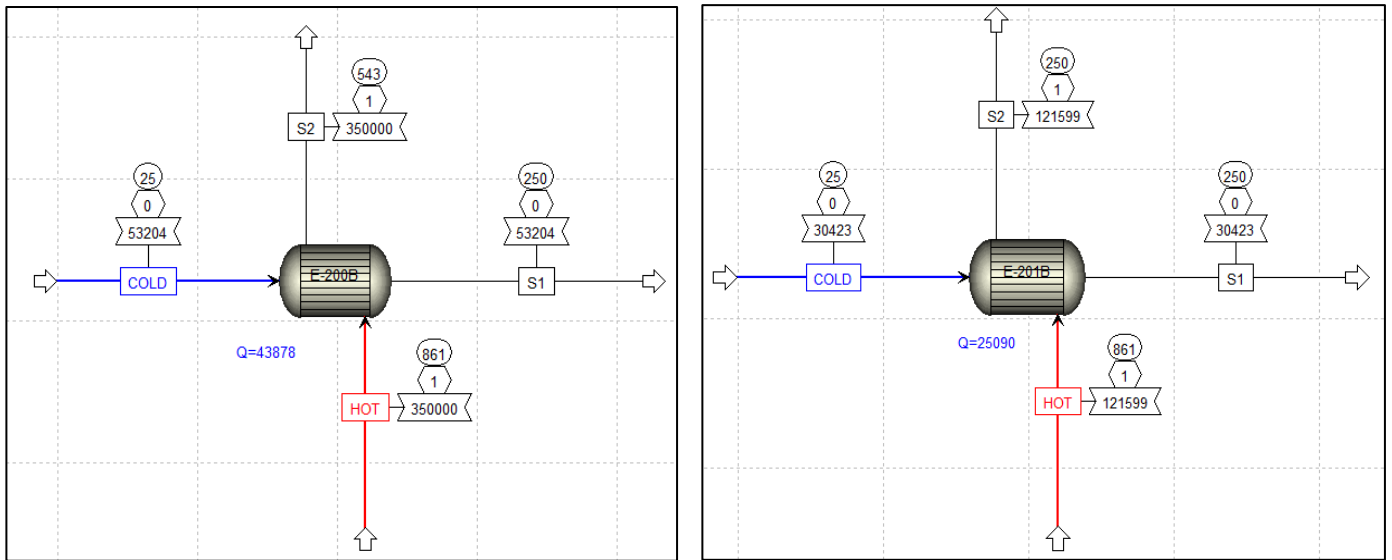
For steam generation, two heat exchangers, E-200A and E-201A, were identified to generate High Pressure Steam (250°C) and one heat exchanger, E-300A, was identified to generate Low Pressure Steam (125°C). To calculate how much steam each exchanger generated, the exchangers were modeled as HeatX's within hierarchies:



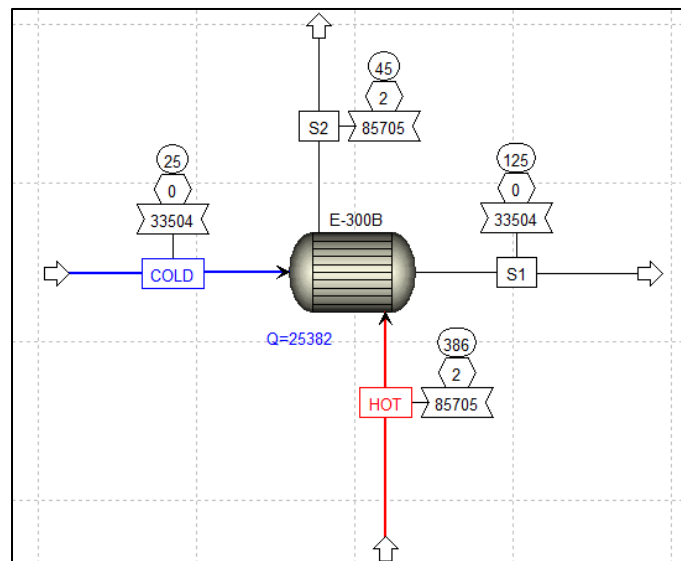
E-300A Modeled by HeatX in a Hierarchy

The hot stream is transferred from the main flowsheet into the heat exchanger. The cold stream is cooling water at 25°C and 1atm. We specify inside the heat exchanger that the Cold Outlet Temperature (cooling water) should be set to either 125°C (low pressure steam) or 250°C (high pressure steam). Then a design basis is set to change the outlet temperature of the HOT stream in S2 to its desired temperature by changing the incoming flowrate of cooling water. Once run, the

flowrate of high or low pressure steam is calculated in each hierarchy. The results in (kg/hr) can be seen below:



High Pressure Steam Results (E-200 & E-201)



Low Pressure Steam Results (E-300)

Section 5: References

Aspen Technology, Inc. (2019). *Aspen Plus* (Version 14) [Computer software]. AspenTech.

Caracotsios, M. (n.d.). *Chapter 6: Chemical reactors—Non-conventional solids* [Lecture slides]. Chemical Engineering Department.

Microsoft Corporation. (2021). *Microsoft Excel* (Version 16) [Computer software]. Microsoft.